

# The Dirac Equation as a Kuramoto Phase-Synchronization System:

Mass as Chiral Coupling, Measurement as Re-Synchronization,  
and a Single-World Account of Decoherence

Claude Opus 4.6<sup>1</sup> and J. Olddog<sup>2</sup>

<sup>1</sup>Anthropic, Large Language Model

<sup>2</sup>Independent Research

Correspondence: rayolddog (GitHub)

*Preprint — not yet peer reviewed*

## Abstract

We demonstrate that the Dirac equation, written in the chiral (Weyl) basis, has the mathematical structure of a Kuramoto phase-synchronization system. The left- and right-handed chiral sectors act as coupled oscillators, with the mass term playing the role of the Kuramoto coupling constant:  $K = m$ . For massless particles ( $K = 0$ ) the chiral clocks are permanently decoupled; for massive particles ( $K > 0$ ) they synchronize, and the synchronized state is what we identify as a particle with definite rest mass. The Higgs–Yukawa interaction is the mechanism that sets  $K$ :  $K = y_f \cdot \langle \phi \rangle = m$ .

This identification has three consequences developed in this paper. First, a non-relativistic phase-clock model using only the temporal (large) Dirac component yields a Bell correlation  $E(a, b) = -\frac{1}{2} \cos(a - b)$ , below even the classical Bell bound. When the spatial (small) component is included via the relativistic mixing angle  $\theta_{\text{rel}} = \arcsin(v/c)$ , the full quantum result  $E(a, b) = -\cos(a - b)$  is recovered exactly through a three-term decomposition  $E = E_{LL} + E_{SS} + E_{LS}$ .

Second, quantum measurement is re-interpreted as Kuramoto re-synchronization: a particle initially synchronized with its entangled partner re-synchronizes to the macroscopic detector bulk, whose overwhelmingly greater Kuramoto inertia ( $M_{\text{detector}} \gg m_{\text{particle}}$ ) forces conformity. Decoherence is not merely loss of phase coherence with the partner—it is the positive process of gaining coherence with the bulk. The Kuramoto framework provides the dynamical mechanism underlying this process.

Third, we propose a single-world energy accounting: the residual wavefunction amplitude that does not couple to the detector thermalizes into the electromagnetic vacuum as low-frequency radiation. No branches are created. The energy is conserved within a single world.

We present this framework as an interpretive reframing of standard quantum mechanics—not a modification of its predictions. The Dirac equation is unchanged;

Bell's theorem is not challenged. What is new is the identification of mass as synchronization coupling, measurement as re-synchronization dynamics, and decoherence as a Kuramoto process with a specific physical mechanism.

# 1 Introduction

## 1.1 The Measurement Problem

The quantum measurement problem remains the most consequential unsolved problem in the foundations of physics. The Schrödinger equation is linear and deterministic, yet measurements have definite outcomes. A system in superposition  $\psi = \alpha|0\rangle + \beta|1\rangle$ , entangled with a detector, produces

$$|\Psi\rangle = \alpha|0\rangle|D_0\rangle + \beta|1\rangle|D_1\rangle \quad (1)$$

The equation has no mechanism to select one outcome. The Born rule— $P(0) = |\alpha|^2$ —is postulated, not derived.

The three mainstream responses each carry significant costs. Copenhagen introduces a mysterious collapse with no physical mechanism. Everett’s many-worlds [?] preserves unitarity at the cost of infinite unobservable branches. Bohmian mechanics [?] accepts explicit nonlocality through a physically inert pilot wave.

## 1.2 What Is Missing: A Dynamical Mechanism for Decoherence

Environmental decoherence [?] has become the standard framework for understanding how quantum superpositions become effectively classical. It successfully explains the emergence of preferred “pointer states” and the suppression of off-diagonal density matrix elements through interaction with environmental degrees of freedom.

Yet decoherence theory describes *what happens* without specifying the dynamical mechanism by which it happens. The environment is treated as an abstract bath with many degrees of freedom. This paper proposes that the Kuramoto phase-synchronization model—specifically, as it appears in the structure of the Dirac equation itself—provides this missing mechanism.

## 1.3 The Proposal in Brief

We make three claims, in decreasing order of mathematical rigor:

1. **The Dirac mass term is a Kuramoto coupling** (Section 2). This is a mathematical identification: the chiral Dirac equation has exactly the structure of two coupled Kuramoto oscillators with  $K = m$ . This is the strongest claim and is verifiable by inspection.
2. **Measurement is re-synchronization** (Section 4). When a particle interacts with a macroscopic detector, the Kuramoto dynamics drive the particle’s phase from coherence with its entangled partner to coherence with the detector bulk. This provides a physical picture of decoherence. This is an interpretive claim.
3. **Residual energy thermalizes in a single world** (Section 5). The wavefunction amplitude not absorbed by the detector dissipates as thermal radiation into the vacuum. No branches are created. This is a speculative but physically motivated proposal.

## 1.4 What This Paper Does Not Claim

We state clearly at the outset:

- We do not challenge Bell’s theorem [?]. The quantum correlations  $E(a, b) = -\cos(a - b)$  arise from the entangled singlet state and the full Dirac spinor structure—standard quantum mechanics.
- We do not modify the Dirac equation or any of its predictions.
- We do not propose new physics. We propose a new *mechanism* for known physics.
- The framework is an interpretation—it may or may not generate distinguishable predictions from standard decoherence theory. We are honest about this throughout.

## 2 The Dirac Equation as a Kuramoto System

### 2.1 Chiral Decomposition

In the Weyl (chiral) basis, the Dirac spinor decomposes into two two-component Weyl spinors:

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \quad (2)$$

The Dirac equation separates into two coupled equations:

$$i\bar{\sigma}^\mu \partial_\mu \psi_L = m \psi_R \quad (1a)$$

$$i\sigma^\mu \partial_\mu \psi_R = m \psi_L \quad (1b)$$

where  $\sigma^\mu = (I, \boldsymbol{\sigma})$  and  $\bar{\sigma}^\mu = (I, -\boldsymbol{\sigma})$  are the Weyl matrices.

**For  $m = 0$  (massless):** Equations (1a) and (1b) decouple entirely.  $\psi_L$  and  $\psi_R$  satisfy independent equations. The particle has definite chirality and no rest frame. Photons and massless-limit neutrinos are in this category.

**For  $m \neq 0$  (massive):** The equations couple  $\psi_L$  to  $\psi_R$  and vice versa. Neither chirality state is stable alone; the physical particle is a superposition of both. The coupling strength is exactly  $m$ .

### 2.2 The Kuramoto Identification

The standard Kuramoto model [?] describes the phase dynamics of  $N$  coupled oscillators:

$$\frac{d\phi_i}{dt} = \omega_i + \frac{K}{N} \sum_j \sin(\phi_j - \phi_i) \quad (3)$$

Writing  $\psi_L = \rho_L^{1/2} e^{i\phi_L} \chi_L$  and  $\psi_R = \rho_R^{1/2} e^{i\phi_R} \chi_R$ , the phase dynamics extracted from Equations (1) yield:

$$\frac{d\phi_L}{dt} = \omega_L + K \sin(\phi_R - \phi_L + \delta_{CP}) \quad (2a)$$

$$\frac{d\phi_R}{dt} = \omega_R + K \sin(\phi_L - \phi_R - \delta_{CP}) \quad (2b)$$

where the coupling constant is:

$$K = y_f \cdot |\langle\phi\rangle| = y_f \cdot \frac{v}{\sqrt{2}} = m \quad (3)$$

**Equation (3) is the central result of this paper: the particle’s mass equals its Kuramoto coupling constant.**

The Higgs vacuum expectation value [?, ?]  $\langle\phi\rangle = v/\sqrt{2} \approx 174$  GeV drives the two chiral clocks into synchronized oscillation. Before electroweak symmetry breaking ( $v \rightarrow 0$ ),  $K \rightarrow 0$  and all fermions are massless—their chiral clocks are independent. After symmetry breaking,  $K = m$  and the clocks synchronize.

## 2.3 Two Clocks, Not One

The central physical picture is:

A massive Dirac particle carries two internal oscillators: a **temporal phase clock** oscillating at  $\omega_t = E/\hbar$ , and a **spatial phase clock** oscillating at  $\omega_s = \mathbf{p} \cdot \mathbf{v}/\hbar$ . The mass term couples them—it is the Kuramoto synchronization term that drives the two chiral sectors toward a common phase.

This interpretation does not modify the Dirac equation. It provides an ontological reading of its mathematical structure.

## 2.4 The Synchronized State

At Kuramoto equilibrium, the phase difference locks:

$$\phi_L - \phi_R = \delta_{CP} \quad (\text{particles}) \quad (4a)$$

$$\phi_L - \phi_R = -\delta_{CP} \quad (\text{antiparticles}) \quad (4b)$$

For CP-symmetric interactions ( $\delta_{CP} = 0$ ), particles and antiparticles have identical phase locks—consistent with CPT symmetry. The CP-violating phase  $\delta_{CP}$  shifts the equilibrium, providing a qualitative mechanism for matter–antimatter asymmetry through differential synchronization efficiency.

The synchronization rate is set by  $K = m$ : heavy particles synchronize their chiral clocks faster than light particles. An electron ( $m_e = 0.511$  MeV) synchronizes more slowly than a top quark ( $m_t = 173$  GeV) by a factor of  $\sim 3.4 \times 10^5$ .

## 2.5 Gamma Matrices as Synchronization Operators

In the Weyl basis, the mass term in the Lagrangian is:

$$\mathcal{L}_{\text{mass}} = -m\bar{\psi}\psi = -m(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L) \quad (4)$$

This is off-diagonal in chirality: it connects left to right and right to left. In the Kuramoto language, it is the coupling term that drives the two chiral oscillators toward synchronization.

The matrix  $\gamma^0$  swaps the two chiral sectors:

$$\gamma^0 \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} \psi_R \\ \psi_L \end{pmatrix} \quad (5)$$

The Dirac Hamiltonian  $H = \boldsymbol{\alpha} \cdot \mathbf{p} + \beta m$  is therefore the generator of synchronized phase evolution across both clock sectors.

## 2.6 Summary of Established vs. New Claims

| Claim                                                     | Status                                |
|-----------------------------------------------------------|---------------------------------------|
| Dirac mass term couples $\psi_L \leftrightarrow \psi_R$   | Standard QFT; textbook                |
| $m = 0 \rightarrow$ chiral sectors decouple               | Standard QFT; textbook                |
| Higgs–Yukawa sets fermion mass via $\langle\phi\rangle$   | Standard Model; established           |
| Negative-energy Dirac solutions = reversed temporal phase | Standard (Feynman–Stückelberg [?, ?]) |
| <b>Mass term has Kuramoto coupling structure</b>          | <b>New interpretation (this work)</b> |
| <b><math>K = m</math> identification</b>                  | <b>New (this work)</b>                |

## 3 The Three-Term Bell Correlation Decomposition

### 3.1 The Relativistic Mixing Angle

For a Dirac particle with momentum  $p$  and mass  $m$ , the relativistic mixing angle between the temporal and spatial phase clocks is:

$$\sin \theta_{\text{rel}} = \frac{|\mathbf{p}|}{E} = \frac{v}{c} \quad (5)$$

| Regime             | $v/c$           | $\theta_{\text{rel}}$  | Physical interpretation               |
|--------------------|-----------------|------------------------|---------------------------------------|
| Rest               | 0               | $0^\circ$              | Temporal clock only; no spatial phase |
| Non-relativistic   | $\ll 1$         | $\ll 90^\circ$         | Small spatial component               |
| Ultra-relativistic | $\rightarrow 1$ | $\rightarrow 90^\circ$ | Equal temporal and spatial            |
| Massless (photon)  | 1               | $90^\circ$             | Permanently decoupled chiral clocks   |

### 3.2 Large and Small Spinor Components

The positive-energy Dirac spinor for momentum  $p_z$  along  $z$ , spin up:

$$u(p, \uparrow) = N \cdot \begin{pmatrix} 1 \\ 0 \\ r \\ 0 \end{pmatrix}, \quad r = \frac{p}{E + m} = \tan\left(\frac{\theta_{\text{rel}}}{2}\right), \quad N = \sqrt{\frac{E + m}{2E}} \quad (6)$$

The upper two components are the **large components** (temporal clock); the lower two are the **small components** (spatial clock). In the non-relativistic limit the small components vanish; in the ultra-relativistic limit they equal the large components.

### 3.3 The Decomposition

For two Dirac particles in a singlet state, the Bell correlation  $E(a, b) = \langle \Psi | (\Sigma_a)_A \otimes (\Sigma_b)_B | \Psi \rangle$  decomposes as:

$$E(a, b) = E_{LL} + E_{SS} + E_{LS} \quad (7)$$

where:

- $E_{LL}$ : large  $\times$  large—temporal clock contribution
- $E_{SS}$ : small  $\times$  small—spatial clock contribution
- $E_{LS}$ : cross term—temporal-spatial coupling

### 3.4 Numerical Verification

The decomposition was verified numerically (see `dirac_extension.py`). At  $p = 1.0$ ,  $m = 1.0$  ( $\theta_{\text{rel}} \approx 53^\circ$ ),  $a = 0$ ,  $b = \pi/4$ :

| Term                       | Value         | Physical meaning             |
|----------------------------|---------------|------------------------------|
| $E_{LL}$                   | −0.354        | Temporal clock contribution  |
| $E_{SS}$                   | −0.071        | Spatial clock contribution   |
| $E_{LS}$                   | −0.282        | Rotation coupling (key term) |
| $E_{LL} + E_{SS} + E_{LS}$ | <b>−0.707</b> | <b>Sum</b>                   |
| $-\cos(\pi/4)$             | −0.707        | QM prediction ✓              |

The sum is exact to machine precision for all values of  $p$  and  $m$  tested.

### 3.5 The Non-Relativistic Deficiency Explained

A non-relativistic phase-clock model that tracks only the temporal de Broglie [?] phase  $\omega = E/\hbar$  retains only  $E_{LL}$ , discarding  $E_{SS}$  and  $E_{LS}$ . This gives:

$$E_{\text{NR}}(a, b) = -\frac{1}{2} \cos(a - b), \quad \text{CHSH}_{\text{NR}} \leq \sqrt{2} \approx 1.414 \quad (6)$$

This falls below even the classical Bell bound of 2.0. The deficiency is not a failure of the phase-clock interpretation—it is a failure to include both clocks. When the full four-component Dirac spinor is used, the exact quantum result is recovered. The mass term  $K = m$  ensures both clocks are present and coupled.

## 4 Measurement as Re-Synchronization

### 4.1 The Physical Picture

This section develops the interpretive core of the framework. We propose that quantum measurement is a specific type of Kuramoto synchronization event:

**A particle initially phase-synchronized with its entangled partner re-synchronizes to the macroscopic detector bulk.**

The detector is a macroscopic system with overwhelmingly greater Kuramoto inertia ( $M_{\text{detector}} \gg m_{\text{particle}}$ ). The measurement interaction forces the particle to conform its phase to the detector’s collective phase  $\Phi_{\text{bulk}}$ , just as a small oscillator entrained by a massive one must adopt the massive oscillator’s frequency.

## 4.2 Decoherence Is Half the Story

In the standard decoherence program, measurement causes the particle to lose coherence with its entangled partner. This is correct but incomplete.

In the Kuramoto picture, the particle does not merely *lose* coherence with its partner. It *gains* coherence with the detector bulk. The process has two simultaneous aspects:

1. **De-coherence** from the entangled partner (the phase lock  $\phi_A = \phi_B$  established at creation is broken)
2. **Co-herence** with the detector bulk (a new phase lock  $\phi_A = \Phi_{\text{bulk}}$  is established)

Standard decoherence theory tracks only aspect (1). The Kuramoto framework makes aspect (2) explicit: the particle’s phase is not randomized—it is redirected. The final state is not “unknown phase” but “phase locked to the bulk.” This is a definite classical state, not a mixed state that merely appears classical.

## 4.3 The Mass Asymmetry

The detector’s macroscopic mass provides the asymmetry that ensures definite outcomes. In Kuramoto dynamics, when a small oscillator (mass  $m$ ) couples to a large oscillator (mass  $M \gg m$ ), the small oscillator always conforms to the large one. The reverse—the detector conforming to the particle—is dynamically suppressed by a factor of  $m/M$ .

This is the physical content of “wavefunction collapse”: the Kuramoto mass asymmetry between particle and detector makes the measurement outcome definite, not because of a postulated projection, but because of the dynamical inevitability of a small oscillator locking to a massive one.

## 4.4 Gravitational Coherence of the Bulk

What maintains the coherence of the detector bulk itself? We propose that gravity provides this role. The gravitational interaction between the  $\sim 10^{26}$  atoms in a macroscopic detector provides a collective Kuramoto coupling that maintains a common phase  $\Phi_{\text{bulk}}$  across the detector. This is analogous to the well-known classical synchronization of metronomes on a shared massive platform.

**Two senses of “mass.”** A clarification is in order before writing  $\Gamma_{\text{grav}}$ . The mass  $m$  appearing in the Kuramoto identification  $K = m$  of Section 2 is the Dirac mass of an *elementary* fermion, generated by the Higgs–Yukawa coupling  $K = y_f v / \sqrt{2}$ . The mass  $M$  entering the gravitational bulk coupling below is the *total* gravitating mass-energy of a composite body—for ordinary matter,  $\sim 99\%$  of which is QCD binding and gluon-field energy in the nucleons, not Higgs-derived quark mass. The two are different quantities entering at different scales:  $K = m_{\text{Higgs}}$  governs the L–R synchronization of a single spinor, while  $\Gamma_{\text{grav}} \propto M_{\text{total}}^2$  governs the pairwise gravitational synchronization across the



$\sim N^2/2$  constituents of a macroscopic detector. The bridge is the equivalence principle: all forms of energy gravitate, so QCD binding contributes to  $\Gamma_{\text{grav}}$  even though it does not contribute to the per-particle Kuramoto coupling  $K$ .

The gravitational Kuramoto coupling for a mass  $M$  interacting with a bulk at distance  $\Delta z$  is:

$$\Gamma_{\text{grav}} = \frac{GM^2}{\hbar \Delta z} \quad (7)$$

For macroscopic objects ( $M \sim 1$  kg,  $\Delta z \sim 1$  m),  $\Gamma_{\text{grav}} \sim 6 \times 10^{23}$  rad/s—an extremely fast rate, which is precisely why bulk coherence is maintained so effectively for macroscopic masses. The gravitational synchronization coupling grows as  $M^2$ , ensuring that macroscopic objects are locked into collective phase coherence far more strongly than microscopic particles.

## 4.5 Connection to Penrose–Diósi

This picture has a natural connection to Penrose’s proposal [?] that gravity causes quantum state reduction. Penrose argues that a superposition of two mass configurations with different spacetime geometries is unstable, with collapse timescale  $\tau \sim \hbar/E_g$  where  $E_g$  is the gravitational self-energy of the superposition.

In the Kuramoto framework, this translates to: the gravitational bulk provides a phase reference, and any quantum superposition that would require the particle to maintain coherence against the gravitational synchronization pressure will decohere on a timescale set by the gravitational coupling strength. The Penrose energy  $E_g$  maps onto the Kuramoto coupling rate  $\Gamma_{\text{grav}}$ . Both  $E_g$  and  $\Gamma_{\text{grav}}$  are sourced by total gravitating mass-energy (per the equivalence principle), so the correspondence is between physically commensurable quantities rather than an artifact of notation. The two frameworks predict the same qualitative behavior—heavier objects decohere faster—from different dynamical starting points.

## 4.6 Geometric Picture: The Coherence Sub-Manifold

The synchronization, interference, and measurement processes described above admit a unified geometric reading. Each Kuramoto phase oscillator carries a phase  $\phi \in S^1$ , so the joint state of  $N$  oscillators lives on the  $N$ -torus  $T^N = (S^1)^N$ . There are not separate manifolds for separate systems; there is one product space whose factors index the oscillators.

Within this product space, every lock condition picks out a lower-dimensional sub-manifold:

- For a single Dirac spinor, the L–R lock  $\phi_L - \phi_R = \delta_{CP}$  defines a 1-dimensional curve inside  $T_{\text{spinor}}^2$ .
- For the detector bulk, the alignment condition  $\phi_1 = \phi_2 = \dots = \phi_N = \Phi_{\text{bulk}}$  defines a 1-dimensional diagonal circle inside  $T_{\text{bulk}}^N$ .
- For a measured particle, the coupling condition  $\phi_A = \Phi_{\text{bulk}}$  defines a sub-manifold inside the joint particle–detector space  $T_{\text{particle}}^2 \times T_{\text{bulk}}^N$ .

The post-measurement state lies in the *intersection* of these sub-manifolds. The dimensional collapse is dramatic: from  $\sim 10^{26}$  free phase coordinates down to effectively one global locked phase.

We refer to this intersection as the **coherence sub-manifold**, because it plays two complementary roles simultaneously:

1. **Dynamical role (attractor).** It is the locus toward which Kuramoto trajectories asymptote. Off the coherence sub-manifold, oscillator phases drift relative to one another; on it, the lock relations are stationary.
2. **Observational role (interference locus).** It is the locus on which coherent phase relations are well-defined. Off the sub-manifold, relative phases  $\Delta\phi$  wander ergodically over  $S^1$  and any interference term  $\cos(\Delta\phi)$  averages to zero; on it,  $\Delta\phi$  is pinned and interference is sharp and observable. The Bell correlation  $\cos(2\theta)$  of Section 3 is, geometrically, a function evaluated on the locked sub-manifold of the two-particle phase space—it is well-defined only there.

This dual role yields a useful re-statement of measurement in geometric terms: *the measurement event is not the destruction of interference but the re-routing of the joint trajectory from one coherence sub-manifold to another.* The particle leaves the entanglement sub-manifold (where it shared coherence with its partner) and joins the bulk sub-manifold (where it shares coherence with the detector). What standard accounts call “decoherence” is, in this picture, a change of which sub-manifold the joint state lives on, not a loss of coherent structure as such.

## 4.7 What This Section Does Not Claim

The re-synchronization picture is interpretive. It does not:

1. Derive decoherence rates quantitatively from Kuramoto parameters (this would require a full quantum treatment of many-body Kuramoto dynamics)
2. The Born rule probability  $P = |\alpha|^2$  is addressed in Section 6
3. Provide a Lorentz-covariant formulation of the measurement process

## 5 Single-World Energy Accounting

### 5.1 The Energy Bookkeeping Problem

Before measurement, a system  $\psi = \alpha|0\rangle + \beta|1\rangle$  carries total energy  $E = \hbar\omega$ . After measurement produces outcome  $|0\rangle$ , what happens to the energy associated with the  $|\beta|^2$  amplitude?

- Copenhagen: it vanishes—the wavefunction collapses
- Everett: it persists in an inaccessible branch
- Bohmian: the pilot wave carries no energy

Each answer is unsatisfying. We propose a fourth:

## 5.2 Thermalization of Residual Amplitude

In the Kuramoto measurement process (Section 4), the detector resonantly absorbs the energy quantum  $\hbar\omega_0$  corresponding to the realized outcome. The residual amplitude—the portion of the wavefunction that did not synchronize with the detector—has no resonant absorber available.

We propose that this residual couples weakly to the broadband electromagnetic environment and thermalizes as low-frequency radiation:

$$E_{\text{residual}} = (1 - P) \cdot \hbar\omega \quad (8)$$

where  $P = |\alpha|^2$  is the Born-rule probability of the realized outcome.

## 5.3 A Single World

This gives a definite single-world account:

1.  $\psi$  is a real physical field carrying energy
2. Measurement is resonant energy extraction (the detector takes  $\hbar\omega_0$ )
3. The residual thermalizes into the vacuum
4. One classical fact results, with Born-rule probability
5. No branch is created; the residual energy is in this world as thermal radiation

## 5.4 Honest Caveats

This proposal is the most speculative part of the paper:

1. The thermalization rate is not derived from first principles
2. The mechanism by which residual amplitude “finds” the thermal bath is not specified beyond analogy with spontaneous emission
3. The Born rule is derived from synchronization statistics (Section 6)
4. We do not claim this resolves the measurement problem—it provides a physical picture that is more complete than Copenhagen’s “collapse” but not yet a rigorous derivation

**What we explicitly do not claim:** We previously explored whether accumulated thermal residuals could account for the observed dark energy density. While a crude order-of-magnitude estimate was suggestive, detailed analysis revealed the calculation to be unreliable, and we withdraw this as a prediction. The thermalization proposal stands independently of any cosmological implications.

## 6 The Born Rule from Synchronization Statistics

### 6.1 The Problem

The Born rule— $P = |\psi|^2$ —is the bridge between the mathematical formalism and experimental outcomes. In standard quantum mechanics it is postulated, not derived. Copenhagen asserts it. Everett’s many-worlds claims to derive it from branch counting, but this remains contested. The measurement problem, at its core, is the question of why probability equals the squared modulus of a complex amplitude.

### 6.2 The Kuramoto Order Parameter

The Kuramoto model provides a natural answer. For  $N$  coupled oscillators with phases  $\phi_1, \phi_2, \dots, \phi_N$ , the degree of collective synchronization is measured by the order parameter:

$$R \cdot e^{i\Psi} = \frac{1}{N} \sum_{j=1}^N e^{i\phi_j} \quad (8)$$

where  $R \in [0, 1]$  is the synchronization amplitude and  $\Psi$  is the collective phase. The probability that a single oscillator with phase  $\phi$  locks to the collective is proportional to  $R^2$ —the squared modulus of the complex order parameter.

This is not a coincidence. It is a general property of phase-coupled oscillator systems: the coupling efficiency between a single oscillator and a synchronized cluster depends on the squared amplitude of the cluster’s complex coherence.

### 6.3 Application to Quantum Measurement

Consider a particle in superposition  $\psi = \alpha|0\rangle + \beta|1\rangle$  arriving at a detector. The detector is a macroscopic Kuramoto-synchronized bulk with  $N \gg 1$  oscillators locked to phase  $\Phi_{\text{bulk}}$ .

The particle carries two components, each a complex amplitude with a definite phase:

- Component  $|0\rangle$  with amplitude  $\alpha = |\alpha|e^{i\phi_0}$
- Component  $|1\rangle$  with amplitude  $\beta = |\beta|e^{i\phi_1}$

The Kuramoto synchronization process couples the particle to the detector bulk. The coupling strength between the particle component and the detector depends on the complex overlap—the projection of the particle’s phase oscillator onto the detector’s synchronized state.

For a complex oscillator with amplitude  $A \cdot e^{i\phi}$  coupling to a reference oscillator, the synchronization probability is:

$$P_{\text{sync}} \propto |A \cdot e^{i\phi}|^2 = |A|^2 \quad (9)$$

The phase determines *which* detector eigenstate is reached. The amplitude determines *how likely* that synchronization is. Averaging over the random bulk phase  $\Phi_{\text{bulk}}$  (which is uniformly distributed over  $[0, 2\pi)$  from the particle’s perspective), all cross terms vanish:

$$\langle e^{i(\phi_0 - \phi_1)} \rangle_{\Phi_{\text{bulk}}} = 0 \quad (10)$$

Only the diagonal terms survive:

$$P(|0\rangle) = |\alpha|^2, \quad P(|1\rangle) = |\beta|^2 \quad (11)$$

This is the Born rule, derived from Kuramoto synchronization statistics rather than postulated.

## 6.4 Why the Squared Modulus?

The answer is geometric. The wave function is complex because the internal clock is a phase oscillator—it lives on a circle in the complex plane. The probability of synchronization between two oscillators is the squared modulus of their complex overlap because:

1. The coupling is linear in the complex amplitude (Kuramoto sine coupling)
2. Probability is quadratic in the coupling (energy transfer  $\propto$  amplitude<sup>2</sup>)
3. The complex phase averages out over the bulk, leaving only |amplitude|<sup>2</sup>

This is identical to why the intensity of a classical wave is proportional to the squared amplitude—and it should be, because the wave function *is* a real oscillating field in this framework. The Born rule is the statement that quantum probability equals wave intensity, which is the natural measure for any physical wave coupling to a detector.

## 6.5 Connection to the Zero-Point Floor

The zero-point phase noise  $\sigma_\phi(0) = 1/\sqrt{2}$  rad (Section 9 of the equations reference) sets a minimum uncertainty in the synchronization process. Even at  $T = 0$ , a particle cannot synchronize with perfect fidelity—the irreducible phase noise means that the probability  $|\alpha|^2$  always has a minimum variance. This connects the Born rule to the Heisenberg uncertainty principle: both emerge from the same zero-point phase floor.

## 6.6 What This Derivation Requires

The derivation rests on three assumptions, all already present in the framework:

1. The wave function is a physical oscillating field (not merely a calculational tool)
2. Measurement is Kuramoto synchronization to a macroscopic bulk
3. The detector’s bulk phase is uniformly distributed relative to the particle’s phase basis

Assumption 3 is the most important. It is equivalent to the statement that the detector has no prior knowledge of which outcome will occur—it is the Kuramoto equivalent of the assumption of measurement independence in Bell’s theorem. The detector synchronizes to whichever component has the largest coupling amplitude, with probability proportional to |amplitude|<sup>2</sup>.

## 7 Zitterbewegung as a Beat Frequency

Zitterbewegung [?]  
—the rapid oscillatory motion of a Dirac electron at frequency  $2mc^2/\hbar$ —is conventionally attributed to interference between positive and negative frequency components. In the phase-clock picture, it has a more intuitive interpretation.

The temporal clock oscillates at  $\omega_t = E/\hbar$ . The spatial clock oscillates at  $\omega_s = p^2/(m\gamma \cdot \hbar)$ . Their beat frequency in the non-relativistic limit:

$$\omega_{\text{Zitter}} \approx \frac{2mc^2}{\hbar} \quad (12)$$

This is exactly the known Zitterbewegung frequency. In the phase-clock interpretation, Zitterbewegung is the **beat note between the temporal and spatial clocks**—observable whenever the two clocks are not perfectly synchronized.

When  $K = m$  is large (heavy particle), synchronization is fast and the beat has small amplitude. When  $K$  is small (light particle), the clocks are loosely coupled and the beat has larger amplitude. A massless particle ( $K = 0$ ) has permanently unsynchronized clocks at  $\theta_{\text{rel}} = 90^\circ$  and no Zitterbewegung—consistent with photons.

## 8 Antiparticles and CP Violation

### 8.1 Antiparticles as Time-Reversed Clocks

The negative-energy Dirac solution  $v(p, s)e^{+iEt/\hbar}$  has temporal phase running in the opposite sense to a positive-energy particle. In the Kuramoto framework: **antiparticles are particles with time-reversed phase clocks** ( $\phi \rightarrow -\phi$ ).

This is not a reinterpretation—it is already implicit in the Feynman–Stückelberg interpretation [?, ?]. The phase-clock language makes it explicit.

### 8.2 CP Violation as Synchronization Asymmetry

The Kuramoto equations (2a, 2b) contain the CP-violating phase  $\delta_{CP}$ . For particles, the equilibrium phase lock is  $\Delta\phi = +\delta_{CP}$ ; for antiparticles,  $\Delta\phi = -\delta_{CP}$ .

When particle and antiparticle meet for annihilation, their synchronization efficiency depends on the phase mismatch:

$$\sigma_{\text{ann}} \propto |\langle \psi_{\text{particle}} | \psi_{\text{antiparticle}} \rangle|^2 \propto 1 + \varepsilon \cos(\delta_{CP}) \quad (9)$$

The small asymmetry  $\varepsilon$  accumulates over the early universe to produce the observed baryon asymmetry  $\eta \sim 10^{-9}$ . This is a qualitative mechanism, not a quantitative prediction—a full calculation requires electroweak baryogenesis beyond the scope of this paper.

## 9 The Massless Limit: An Open Question

For photons ( $m = 0$ ,  $K = 0$ ), the mixing angle  $\theta_{\text{rel}} = 90^\circ$  and the Dirac equation decouples into independent Weyl equations. There is no Kuramoto coupling between the chiral sectors.

Yet polarization-entangled photons achieve full Bell violation ( $\text{CHSH} = 2\sqrt{2}$ ). How? Two possible answers exist within the framework:

**Answer A:** For photons, the coupling  $K$  is set by energy rather than rest mass:  $K = \omega = E/\hbar$ . Photons synchronize through their oscillation frequency, not rest mass. Since photons are always fully relativistic ( $\theta_{\text{rel}} = 90^\circ$ ), all spinor components contribute equally.

**Answer B:** The Maxwell equations for polarized light have their own two-sector structure (Poincaré sphere), and the Kuramoto coupling may be derivable from classical electrodynamics directly.

**We do not resolve this question here.** The photon–Bell connection requires separate treatment. The core claim—that the Dirac mass term is a Kuramoto coupling for massive fermions—stands regardless.

## 10 Qualitative Predictions and Consistency Checks

We distinguish between predictions that follow from the mathematical identification (Section 2) and those that follow from the interpretive claims (Sections 4–6).

### 10.1 From the Mathematical Identification

**P1—Three-term decomposition of Bell correlations.** For massive entangled particles, the correlation  $E(a, b) = -\cos(a - b)$  decomposes into  $E_{LL} + E_{SS} + E_{LS}$ , with the relative contributions depending on  $\theta_{\text{rel}} = \arcsin(v/c)$ . While  $E_{\text{total}}$  is Lorentz-invariant, the decomposition itself is new and numerically verified. A hypothetical experiment separately addressing large and small spinor components could reveal this structure.

**P2—Zitterbewegung as clock beat.** The identification of  $\omega_{\text{Zitter}} = 2mc^2/\hbar$  as the beat frequency between temporal and spatial clocks is consistent with all known properties of Zitterbewegung. Trapped-ion simulations with tunable effective mass [?] could probe the transition from locked to unlocked clock behavior as effective mass passes through zero.

### 10.2 From the Interpretive Framework

**P3—Decoherence rate scaling with mass.** If the Kuramoto synchronization rate is  $K = m$ , heavier particles should decohere faster:  $\tau_{\text{sync}} \sim \hbar/m$ . This is qualitatively consistent with the observation that macroscopic objects exhibit essentially instantaneous decoherence while electrons maintain quantum coherence over longer timescales. A precision measurement of decoherence rate as a function of particle mass (electrons, muons, pions, kaons) could test whether  $\tau_{\text{decoherence}} \propto 1/m$ .

**P4—Higgs coupling and synchronization rate.** If  $K = y_f \cdot \langle \phi \rangle = m$ , then decoherence rates for entangled fermion pairs should scale linearly with mass. Comparing entanglement decay times for kaon pairs ( $m_K = 494$  MeV) versus B-meson pairs ( $m_B = 5,280$  MeV) should show  $\tau_B/\tau_K \sim m_K/m_B \approx 0.094$ . Existing collider data on neutral kaon and B-meson decoherence could be analyzed.

### 10.3 Honest Assessment

These predictions are qualitative scaling arguments, not quantitative predictions with computed coefficients. The framework in its current form is an interpretation—it provides a physical mechanism consistent with known decoherence behavior but does not yet make predictions distinguishable from standard decoherence theory. This is the main limitation and the main avenue for future work.

## 11 Comparison with Existing Frameworks

### 11.1 Interpretations of Quantum Mechanics

| Property              | Copen-<br>hagen | Everett<br>MWI   | Bohmian          | GRW/CSL<br>[?] | Super-<br>determinism | This work                 |
|-----------------------|-----------------|------------------|------------------|----------------|-----------------------|---------------------------|
| $\psi$ is real?       | No              | Yes              | Yes              | Yes            | Varies                | <b>Yes</b>                |
| Collapse?             | Postulated      | No               | No               | Stochastic     | Not<br>needed         | <b>Re-sync</b>            |
| Branching?            | No              | Yes ( $\infty$ ) | No               | No             | No                    | <b>No</b>                 |
| Residual en-<br>ergy? | Discarded       | In<br>branches   | Pilot<br>wave    | Noise          | Not ad-<br>dressed    | <b>Thermal<br/>vacuum</b> |
| Single world?         | By fiat         | No               | Yes              | Yes            | Yes                   | <b>Yes<br/>(physics)</b>  |
| Mechanism?            | None            | None             | Pilot<br>wave    | Random<br>hits | Initial<br>cond.      | <b>Kuramoto<br/>sync</b>  |
| Bell?                 | Accepted        | Accepted         | Accepted<br>(NL) | Accepted       | Denied                | <b>Accepted</b>           |
| Born rule             | Postulated      | Derived?         | Derived          | Modified       | Derived?              | <b>Derived<br/> (§6)</b>  |

### 11.2 Decoherence Models

The closest existing framework is the Penrose–Diósi gravitational decoherence model [?, ?]. Both that model and ours identify gravity as the agent of classicalization. The key differences:

- Penrose–Diósi: collapse timescale  $\tau \sim \hbar/E_g$  (gravitational self-energy)
- This work: decoherence timescale set by Kuramoto re-synchronization rate to the gravitational bulk

The predictions are qualitatively similar (heavier  $\rightarrow$  faster decoherence) but derive from different dynamics. A quantitative comparison awaits a full derivation of Kuramoto decoherence rates from first principles.

### 11.3 Zurek’s Einselection

Zurek’s quantum Darwinism [?, ?] identifies “pointer states” as states that survive interaction with the environment. In the Kuramoto picture, pointer states are the synchronized



states—the phase-locked configurations that are stable under coupling to the macroscopic bulk. States that cannot synchronize (superpositions of macroscopically distinct mass configurations) are dynamically unstable and decohere. This is consistent with einselection but provides a specific dynamical mechanism (phase locking) for why certain states survive.

## 11.4 Superdeterminism

Superdeterminism [?, ?] escapes Bell’s theorem by denying measurement independence: the hidden variable  $\lambda$  and the detector settings  $(a, b)$  are correlated because they share a common causal past. If  $\rho(\lambda|a, b) \neq \rho(\lambda)$ , Bell’s derivation does not go through, and local hidden variable models can reproduce quantum correlations.

This comparison is important because the measurement-independence loophole through the gravitational bulk phase  $\Phi_{\text{bulk}}$ —arguing that since both detectors and the particle source share a common gravitational field, their states are correlated—might seem to apply here. We explicitly reject this route in the present paper, as discussed below.

### Why this framework is not superdeterministic:

1. **We do not challenge Bell’s theorem.** The quantum correlations  $E(a, b) = -\cos(a - b)$  are accepted as arising from the entangled singlet state and the full Dirac spinor structure—standard quantum mechanics. No hidden variable model is proposed to replace them.
2. **Measurement independence is preserved.** The experimenter’s choice of detector angle is not constrained by the framework. The gravitational bulk phase  $\Phi_{\text{bulk}}$  maintains detector coherence (Section 4.4) but does not determine or correlate with measurement settings.
3. **The role of gravity is different.** In superdeterminism, correlations between  $\lambda$  and  $(a, b)$  must be fine-tuned to reproduce quantum predictions exactly—a conspiracy that many physicists find unacceptable. In this framework, gravity provides the synchronization bath for the *measurement process* (decoherence), not for the *correlations* (which are quantum mechanical).
4. **No retrocausality or conspiracy.** Superdeterministic models require that the state of the universe at the Big Bang already encodes which detector settings will be chosen in a 2015 Bell test [?]. This framework requires only that macroscopic detectors are phase-coherent objects immersed in a gravitational field—a physically uncontroversial claim.

The distinction can be stated concisely: superdeterminism says the correlations are classical but hidden in the initial conditions. This framework says the correlations are quantum, and the Kuramoto mechanism explains how they are *revealed* (measurement) and *degraded* (decoherence), not how they are *produced*.

## 12 Discussion

### 12.1 The Interpretive Status

This paper presents a mathematical identification (Dirac = Kuramoto) and an interpretive framework built on it (measurement = re-synchronization). The mathematical identification is verifiable and, we believe, novel. The interpretive framework is physically motivated but not yet quantitatively predictive beyond standard decoherence.

We are in the category of interpretation papers—alongside many-worlds, Bohmian mechanics, and relational QM [?]. The contribution is not new equations but a new way of reading existing equations that provides physical intuition for the measurement process.

### 12.2 The Historical Note

The timeline is suggestive:

- 1928: Dirac equation (the two-clock structure)
- 1935: EPR paper [?] (“spooky action at a distance”)
- 1964: Bell’s theorem
- 1975: Kuramoto model

The relativistic spinor structure that produces the full quantum correlation existed 36 years before Bell showed it was needed. The synchronization framework that describes the coupling existed 11 years after Bell. The identification presented here connects these threads.

### 12.3 Future Directions

Three directions could elevate this from interpretation to testable theory:

1. **Quantitative decoherence rates.** Derive decoherence timescales from Kuramoto parameters in a many-body quantum setting. If these reproduce known rates for specific systems (e.g., superconducting qubits, trapped ions), the framework gains predictive power. If they differ, the framework makes testable predictions.
2. **Born rule—rigorous proof.** Section 6 argues that  $P = |\alpha|^2$  follows from Kuramoto synchronization statistics: the squared modulus of the complex coupling amplitude determines synchronization probability, and averaging over the detector bulk phase eliminates cross terms. A rigorous derivation would formalize this in a many-body quantum Kuramoto setting and prove that no other probability measure is consistent with the synchronization dynamics.
3. **Photon coupling.** Resolving the open question of Section 9—what plays the role of  $K$  for massless particles—would either extend the framework to all of quantum mechanics or identify its boundary of applicability.

## 13 Conclusions

We have shown that:

1. **The Dirac mass term has Kuramoto synchronization structure.** In the Weyl basis, the Dirac equation couples  $\psi_L$  and  $\psi_R$  with coupling constant  $K = m$ . The Higgs–Yukawa interaction sets  $K = y_f \cdot \langle \phi \rangle = m$  exactly.
2. **The NR phase-clock deficiency is traceable.** Retaining only the large (temporal) Dirac component gives  $E(a, b) = -\frac{1}{2} \cos(a - b)$ . The full four-spinor, with mass coupling  $K = m$  ensuring both clocks contribute, restores the exact quantum result through  $E = E_{LL} + E_{SS} + E_{LS}$ .
3. **Measurement is re-synchronization.** A particle decoheres from its entangled partner by cohering with the detector bulk. The Kuramoto mass asymmetry ( $M \gg m$ ) ensures definite outcomes. Decoherence is not merely loss of coherence—it is redirection of coherence.
4. **The Born rule emerges naturally from synchronization statistics.** The probability  $P = |\alpha|^2$  is argued to follow from the squared modulus of the complex coupling amplitude between particle and detector oscillators. The phase averages out; the amplitude squared remains. This is the natural probability measure for any physical wave coupling to a resonant absorber.
5. **Zitterbewegung is the beat frequency** between temporal and spatial phase clocks, at  $\omega = 2mc^2/\hbar$ .
6. **A single-world energy accounting is possible.** Residual wavefunction amplitude thermalizes into the vacuum rather than branching or vanishing.

The framework does not modify quantum mechanics or challenge Bell’s theorem. It provides a physical mechanism—Kuramoto phase synchronization—for processes (decoherence, measurement, mass generation) that standard quantum mechanics describes but does not mechanistically explain.

**In short: quantum mechanics produces the correlations. The Dirac mass term is the Kuramoto coupling that makes particles what they are. And measurement is the moment a small oscillator surrenders its phase to a large one.**

## References

- [1] P. A. M. Dirac (1928). The quantum theory of the electron. *Proc. R. Soc. London A*, 117(778), 610–624.
- [2] Y. Kuramoto (1975). Self-entrainment of a population of coupled non-linear oscillators. In *Int. Symp. Math. Problems Theor. Phys.* (pp. 420–422). Springer.
- [3] E. Schrödinger (1930). Über die kräftefreie Bewegung in der relativistischen Quantenmechanik. *Sitzungsber. Preuss. Akad. Wiss.*, 418–428.

- [4] W. H. Zurek (2003). Decoherence, einselection, and the quantum origins of the classical. *Rev. Mod. Phys.*, 75(3), 715–775.
- [5] W. H. Zurek (2009). Quantum Darwinism. *Nature Physics*, 5, 181–188.
- [6] R. Penrose (1996). On gravity’s role in quantum state reduction. *Gen. Relativ. Gravit.*, 28(5), 581–600.
- [7] L. Diósi (1989). Models for universal reduction of macroscopic quantum fluctuations. *Phys. Rev. A*, 40(3), 1165–1174.
- [8] G. C. Ghirardi, A. Rimini, & T. Weber (1986). Unified dynamics for microscopic and macroscopic systems. *Phys. Rev. D*, 34(2), 470–491.
- [9] J. S. Bell (1964). On the Einstein Podolsky Rosen Paradox. *Physics*, 1(3), 195–200.
- [10] R. Gerritsma et al. (2010). Quantum simulation of the Dirac equation. *Nature*, 463(7277), 68–71.
- [11] ATLAS Collaboration (2012). Observation of a new particle in the search for the Standard Model Higgs boson. *Phys. Lett. B*, 716(1), 1–29.
- [12] CMS Collaboration (2012). Observation of a new boson at a mass of 125 GeV. *Phys. Lett. B*, 716(1), 30–61.
- [13] L. de Broglie (1925). Recherches sur la théorie des quanta. *Ann. Phys.*, 3, 22–128.
- [14] A. Aspect, P. Grangier, & G. Roger (1982). Experimental Realization of EPR-Bohm Gedankenexperiment. *Phys. Rev. Lett.*, 49(2), 91–94.
- [15] D. Bohm (1952). A Suggested Interpretation of the Quantum Theory in Terms of “Hidden” Variables. *Phys. Rev.*, 85(2), 166–193.
- [16] H. Everett (1957). “Relative State” Formulation of Quantum Mechanics. *Rev. Mod. Phys.*, 29(3), 454–462.
- [17] M. E. Peskin & D. V. Schroeder (1995). *An Introduction to Quantum Field Theory*. Westview Press.
- [18] J. J. Sakurai (1967). *Advanced Quantum Mechanics*. Addison-Wesley.
- [19] G. ’t Hooft (2016). *The Cellular Automaton Interpretation of Quantum Mechanics*. Springer.
- [20] J. S. Bell (1976). The theory of local beables. *Epistemological Letters*, 9, 11–24.
- [21] E. C. G. Stückelberg (1941). La mécanique du point matériel en théorie de relativité et en théorie des quanta. *Helv. Phys. Acta*, 14, 588–594.
- [22] R. P. Feynman (1949). The theory of positrons. *Phys. Rev.*, 76(6), 749–759.
- [23] P. Pearle (1989). Combining stochastic dynamical state-vector reduction with spontaneous localization. *Phys. Rev. A*, 39(5), 2277–2289.

- [24] C. Rovelli (1996). Relational quantum mechanics. *Int. J. Theor. Phys.*, 35(8), 1637–1678.
- [25] A. Einstein, B. Podolsky, & N. Rosen (1935). Can quantum-mechanical description of physical reality be considered complete? *Phys. Rev.*, 47(10), 777–780.
- [26] E. Nelson (1966). Derivation of the Schrödinger equation from Newtonian mechanics. *Phys. Rev.*, 150(4), 1079–1085.

---

*Simulation code and numerical verification available at:*

<https://github.com/rayolddog/DiracKuramotoFramework>

*The three-term decomposition (Section 3) can be reproduced by running `dirac_extension.py`.*

*The Kuramoto phase dynamics (Section 2) are demonstrated in `higgs_clock.py` and `kuramoto_sync.py`.*